

Week 4 Lecture Prepare Quiz: Logistic Regression, Stochastic Gradient Descent, Multi-Class Classification

- Due Sep 30 at 11a.m.
- Points 14
- Questions 14
- Available Sep 24 at 12a.m. - Sep 30 at 11a.m.
- Time Limit None
- Allowed Attempts Unlimited

This quiz was locked Sep 30 at 11a.m..

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	6 minutes	14 out of 14

❗ Correct answers are hidden.

Score for this attempt: 14 out of 14

Submitted Sep 24 at 1:40p.m.

This attempt took 6 minutes.



Question 1

1 / 1 pts

Which of the following is the correct formula for the sigmoid activation function $\sigma(z)$?

- ☐ $\sigma(z) = \frac{e^z}{e^z + e^{-z}}$
- ☐ $\sigma(z) = \max(0, z)$
- ☐ $\sigma(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$
- ☒ $\sigma(z) = \frac{1}{1 + e^{-z}}$



Question 2

1 / 1 pts

The cross-entropy loss $\mathcal{L}_{CE}(y, t)$ for logistic regression is defined as which expression?

- ☐ $\mathcal{L}_{CE}(y, t) = t \cdot y + (1 - t) \cdot (1 - y)$
- ☐ $\mathcal{L}_{CE}(y, t) = (y - t)^2$
- ☐ $\mathcal{L}_{CE}(y, t) = |y - t|$
- ☒ $\mathcal{L}_{CE}(y, t) = -t \log(y) - (1 - t) \log(1 - y)$



Question 3

1 / 1 pts

The output $y = \sigma(\mathbf{w}^\top \mathbf{x})$ of a logistic regression model represents which quantity?

- ☐ The Euclidean distance from input $\mathbf{x}$ to the decision boundary
- ☐ The predicted class label assigned to input $\mathbf{x}$
- ☒ The probability of the positive class given input $\mathbf{x}$
- ☐ The magnitude of the weight vector $\mathbf{w}$ projected onto $\mathbf{x}$



Question 4

1 / 1 pts

Why is the squared error loss problematic when applied to binary classification?

- ☐ It requires the model to output continuous values between negative and positive infinity
- ☐ It produces output values that are outside the range $(0, 1)$
- ☐ It assigns infinite loss to every misclassified example
- ☒ It penalizes confident correct predictions



Question 5

1 / 1 pts

Which of the following is an advantage of logistic regression over k -nearest neighbors?

- ☐ Logistic regression can learn complex nonlinear decision boundaries
- ☐ Logistic regression stores the entire training dataset for making predictions
- ☐ Logistic regression automatically increases model complexity with more training data
- ☒ Logistic regression provides fast predictions by using a fixed set of learned weights



Question 6

1 / 1 pts

In full batch gradient descent, the gradient of the cost function is computed using how many training examples per update?

- ☐ A small random subset of the training examples
- ☒ All N training examples in the dataset
- ☐ Only the training examples that are currently misclassified
- ☐ A single randomly selected training example



Question 7

1 / 1 pts

In stochastic gradient descent, a mini-batch is best described as which of the following?

- ☐ The set of all misclassified examples identified after each epoch
- ☐ The entire training dataset used in each gradient computation

- ☐ A single example selected in fixed sequential order from the dataset
- ☒ A small subset of k examples used to estimate the gradient



Question 8

1 / 1 pts

A ravine in the optimization landscape arises when which condition holds?

- ☐ The loss function is equally steep in every weight direction
- ☐ Training progresses smoothly as there is a clear direction to the minimum
- ☒ Different weight dimensions have very different gradient magnitudes
- ☐ The training data contains an unequal number of positive and negative examples



Question 9

1 / 1 pts

If the learning rate is set too large during gradient descent optimization, what is the likely outcome?

- ☒ The optimization diverges and the loss function increases without bound
- ☐ The optimization finds a better minimum with higher generalization accuracy
- ☐ The optimization converges more slowly toward the minimum of the loss
- ☐ The optimization path becomes smoother and more stable over time



Question 10

1 / 1 pts

How can the noisy gradient updates in SGD be beneficial during optimization?

- ☒ They can help the optimization escape flat regions such as plateaus
- ☐ They guarantee that the algorithm always finds the global minimum
- ☐ They completely eliminate the need for feature standardization
- ☐ They ensure convergence in fewer iterations than full batch descent



Question 11

1 / 1 pts

In multi-class classification with K classes, the target is represented as which of the following?

- ☒ A one-hot vector in $\mathbb{R}^K$ with a single 1 and the rest 0 s
- ☐ A binary vector in $\{0, 1\}^K$ with multiple entries equal to 1
- ☐ An integer value from 1 to K stored as a single scalar
- ☐ A probability distribution over the K classes that sums to one



Question 12

1 / 1 pts

The softmax function maps a vector of logits $\mathbf{z} \in \mathbb{R}^K$ to which type of output?

- ☐ A binary vector where the entry for the largest logit becomes **1**
- ☒ A probability distribution
- ☐ A normalized vector with unit Euclidean magnitude
- ☐ A vector of integer rankings from the best class to the worst class



Question 13

1 / 1 pts

The decision boundary between classes i and j in softmax regression is defined by which condition?

- ☐ The cross-entropy loss for class i equals the loss for class j
- ☐ The softmax probability for class i equals exactly **0.5**
- ☒ The logits for both classes are equal
- ☐ The weight vectors for classes i and j are perpendicular to each other



Question 14

1 / 1 pts

For a softmax regression model with K classes, how many pairwise decision boundaries exist?

- ☐ K^2 boundaries
- ☐ K boundaries
- ☒ $K(K - 1)/2$ boundaries
- ☐ $K - 1$ boundaries

Quiz Score: 14 out of 14